Informational Power Through Pivotality: When a Hegemon May Choose Consensus

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Abstract

Why would a powerful state choose consensus when majority rule seems to offer more control? I develop a formal model in which consensus can benefit a hegemon without giving it agenda power. The baseline sets the hegemon's recognition probability to zero: weak states make proposals, while the hegemon is privately informed about its participation threshold. Proposals are fixed-pie institutional packages and weak-state allocations, such as quota shares or flexibility provisions, that are always feasible but costly to the weak coalition. Under majority rule, weak states can avoid screening when the hegemon is not a cheaper coalition partner than a weak voter. Under unanimity, the hegemon is pivotal, so weak proposers must choose between a pooling package, a low-threshold package, and delay. Under a specified passive-belief assessment, the selected Round-1 outcome lies in this candidate set. Unanimity need not globally dominate majority. Rather, weak-state entry under unanimity is nested in entry under majority, and when both rules form the hegemon's ranking is determined by the sign of its payoff gap. In a working OPEC-style calibration, unanimity forms for entry costs up to 0.0619583, majority up to 0.0833333, and the hegemon prefers unanimity for beliefs below 0.7 when both rules form. Consensus can therefore be a technology of informational power through pivotality, not merely a constraint on power.

1 Introduction

Why would a powerful state ever choose consensus? In many international organizations, the rule appears to give weaker members exactly what power politics should deny them: a veto. If the powerful can design institutions, majority rule seems more attractive. It lets coalitions pass agreements without unanimous consent, while consensus gives every participant the ability to block.

This paper offers a different answer. Consensus can favor a powerful privately informed actor not because it gives that actor formal agenda power, but because it transforms its approval into an informational constraint on weaker states. When every agreement requires the hegemon's approval, weak states must decide how generous an institutional package must be without observing the hegemon's true participation threshold. If the low-threshold type would accept a modest package but the high-threshold type requires more, weak states face a screening problem. The rent arises when weak states pool the types by offering enough to secure agreement from both.

The key design choice in the model is to separate three sources of power that are often bundled together: outside payoffs, pivotality, and proposal power. The baseline deliberately sets the hegemon's recognition probability to zero, $\pi_H = 0$. Only weak states are recognized as proposers in the bargaining rounds. The hegemon is powerful because it is privately informed and pivotal under unanimity, not because it controls the agenda. Proposals are relative institutional packages, such as quota shares, production flexibility, monitoring provisions, or enforcement burdens. They are always feasible; their cost is that they reduce the residual surplus available to weak states.

The model produces four results. First, majority rule is a no-screening benchmark only under a substantive condition: the low-threshold hegemon must not be cheaper to buy than a weak voter. In primitives, $t_0 - (1-\beta)o_0 \geq \beta/m$. If this condition fails, majority can also screen the hegemon by using it as a cheap coalition partner. Second, under unanimity, terminal bargaining has a threshold form: weak proposers choose a low-threshold package at low beliefs and a pooling package at higher beliefs. Third, in Round 1, under a complete passive-belief assessment, the selected outcome lies in a three-candidate set: pooling, low-only acceptance, and no-information rejection. The result is explicitly scoped to the stated assessment; it is not a characterization under arbitrary off-path beliefs. Fourth, because majority preserves the weak coalition's full surplus when the no-screening condition holds, weak-state entry under unanimity is nested in entry under majority.

The institutional comparison is therefore a classification, not an unconditional ranking of rules. If neither rule forms, the hegemon is indifferent. If only majority forms, majority is better. If both rules form, the hegemon's ranking is determined by the sign of $\Delta_H(\mu)$, the payoff gap between unanimity and majority. In the working OPEC-style calibration, unanimity forms only when the entry cost per weak state satisfies $\chi \leq 0.0619583$, while majority forms up to $\chi = 0.0833333$. Conditional on both rules forming, the hegemon prefers unanimity for $\mu < 0.7$, is indifferent at $\mu = 0.7$, and prefers majority for higher beliefs.

The substantive implication is that consensus can be an instrument of power even when it withholds formal agenda control from the hegemon. Its value is greatest when weak states need the hegemon's participation but cannot observe the terms that make participation worthwhile. This mechanism is especially relevant in organizations such as OPEC, where members bargain over relative institutional terms and where the pivotal producer's outside opportunities, spare capacity, and costs of restraint are hard for others to verify.

2 Consensus, Information, and Institutional Design

The argument contributes to three literatures. First, it speaks to rational-design accounts of international institutions. Existing work emphasizes how institutional rules help states manage uncertainty, commitment, and distributional conflict (Koremenos, Lipson, and Snidal 2001). Consensus is often interpreted as a device for protecting participants from unfavorable outcomes or for encouraging broad participation (Gould 2016). This paper identifies a different channel: uncertainty can make consensus valuable to the privately informed pivotal actor, not only to the uninformed states seeking insurance.

Second, the paper qualifies self-binding and informal-power accounts. Self-binding theories explain why powerful states may accept constraints to make cooperation credible (Keohane 1984; Ikenberry 2001). Informal-power accounts show how powerful states can dominate outcomes even under formally egalitarian rules (Steinberg 2002; Stone 2011; Gruber 2000). The mechanism here is distinct from both. The hegemon does not need to be trusted as a self-bound leader, and it does not need to control drafting or side payments. It benefits because unanimity makes its privately known participation threshold pivotal.

Third, the model builds on bargaining theories in which proposal rights and voting rules shape distributive outcomes (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). The departure is the weak-state agenda baseline. By setting $\pi_H = 0$, the model removes formal proposal power from the hegemon and asks whether pivotality alone can create informational rents. The answer is yes, but only in the classified regions where the screening rent exceeds the majority benchmark. This also connects the paper to work on information and veto bargaining (Bardhi and Guo 2018; Kim, Kim, and Van Weelden 2025), while shifting attention from persuasion by an information designer to institutional rules that determine whether private thresholds matter.

3 A Motivating Example

Consider three states: a hegemon H and two weak states. The weak coalition has a fixed institutional surplus normalized to one. A proposal is a package $y \in [0, 1]$. A higher y is better for H and costlier for the weak coalition. The hegemon privately knows its acceptance threshold. The low type accepts any package $y \geq t_0$, the high type accepts any package $y \geq t_1$, and $0 < t_0 < t_1 < 1$. Weak states know the thresholds but not the type.

Under majority rule, the two weak states can pass a package without H. They set $y = 0$, divide the weak surplus among themselves, and leave H to its external payoff. The vote of H can be recorded, but it is not pivotal on this path. The private threshold of H therefore creates no bargaining problem for the weak coalition.

Under unanimity, no package passes without H's approval. A weak proposer can offer $y = t_0$, which is accepted only by the low type, or $y = t_1$, which is accepted by both types. The low-threshold package gives the proposer payoff $(1 - \mu)(1 - t_0)$, where μ is the probability that H is high threshold. The pooling package gives payoff $1 - t_1$. The weak proposer switches to the pooling package when

$$\mu > \frac{t_1 - t_0}{1 - t_0}.$$

The informational rent is simple. When the pooling package is used, the low type receives t_1 even though it would have accepted t_0. This overpayment is not a mistake; it is the price of making a privately informed veto player willing to participate. The general model keeps this fixed-pie logic, adds two bargaining rounds, collective entry by weak states, and a majority benchmark that clarifies when majority truly removes screening.

4 The Model

Definition 1 (Fixed-pie relative-package institutional design game)**.** *There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states $W_1, \dots, W_m$. Nature draws $\theta \in \{0, 1\}$. The hegemon observes θ; weak states share prior $\mu = \Pr(\theta = 1)$. The weak coalition's institutional surplus is fixed and normalized to one.*

In each bargaining round, a recognized weak proposer W_p chooses a proposal

$$(y, x_1, \dots, x_m)$$

with

$$y \in [0, \bar{y}], \qquad x_i \geq 0, \qquad y + \sum_{i=1}^{m} x_i \leq 1.$$

The component y is a relative institutional package favoring H. A higher y is more favorable to H and reduces the residual payoff available to weak states one-for-one. If the proposal passes, weak state W_i receives x_i, and any slack $1 - y - \sum_i x_i$ is assigned to the weak proposer. Thus the proposer can keep the residual and can discriminate payments among weak voters. The objects c_M, $c(\mu)$, and $c(0)$ below are the minimum weak-voter allocations needed to make the relevant weak voters willing to approve.

The hegemon's terminal net participation threshold is t_θ. If type θ accepts package y, its current payoff is

$$u_H^\theta(\textit{accept } y) = o_\theta + y - t_\theta,$$

where o_θ is the payoff type θ receives when no agreement includes H. In a terminal round, this is equivalent to accepting iff $y \geq t_\theta$. In Round 1, acceptance is dynamic: if rejection induces posterior ν, type θ compares current acceptance with $\beta C_\theta(\nu)$, its discounted continuation payoff. Hence the

Round-1 acceptance threshold is

$$a_\theta(\nu) = t_\theta - o_\theta + \beta C_\theta(\nu). \qquad \text{(Dynamic Acceptance)}$$

The maintained terminal Threshold Order is

$$0 \le t_0 < t_1 \le \bar{y} \le 1. \qquad \text{(Threshold Order)}$$

There are two bargaining rounds and a common discount factor $\beta \in (0, 1]$. The baseline recognition protocol sets $\pi_H = 0$ in both rounds: only weak states are recognized as proposers, each with probability $1/m$. Under unanimity, all N states must vote yes. Under majority, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ yes votes. The proposer is counted as voting yes; all other states vote simultaneously; the full vote vector is public after the ballot.

Under majority, if more than one minimum-cost weak coalition can pass the proposal, the recognized proposer selects the required weak voters symmetrically among the non-proposing weak states. Recognition is uniform and weak states are ex ante symmetric. Thus ex post payments can differ across weak states, but the representative ex ante weak-state payoff equals total weak payoff divided by m.

Weak states decide collectively whether to enter. Entry is all-or-nothing and costs χ per weak state. The institution forms under rule $R \in \{U, M\}$ if the representative ex ante weak-state payoff under that rule is at least χ. Because recognition and coalition selection are symmetric, this is equivalent to requiring each weak state's expected individual payoff to be at least χ.

Definition 2 (Passive-belief assessment)**.** *The baseline uses a pure-strategy assessment with the following belief discipline. First, unilateral deviations by weak voters do not update beliefs about θ, because weak states do not observe H's type. Second, when H's prescribed vote separates types, the posterior is pinned down by H's vote. Third, when both types of H are prescribed to vote yes and H instead votes no, the posterior after failure is $\nu = 1$. Fourth, when both types of H are prescribed to vote no and H instead votes yes, the posterior after failure is $\nu = 0$. Fifth, as an exception for proposals designed to fail by a designated weak voter's rejection, H's nonpivotal vote is not used as a signal: the posterior remains the prior μ after any vote by H.*

Players approve when their vote is pivotal and approval weakly beats rejection. If a weak proposer is indifferent among payoff-maximizing proposals, the baseline selects the proposal that minimizes H's expected payoff at the current belief. This tie-break operates only among candidates already admissible under the assessment; it does not resolve multiplicity created by unrestricted off-path beliefs.

The passive-belief assessment is a maintained refinement, not a claim about unrestricted PBE. Standard PBE leaves beliefs after some off-path histories underdetermined; the paper's formal results characterize the selected outcome under Definition 2 (Kreps and Wilson 1982; Fudenberg and Tirole 1991; Cho and Kreps 1987).

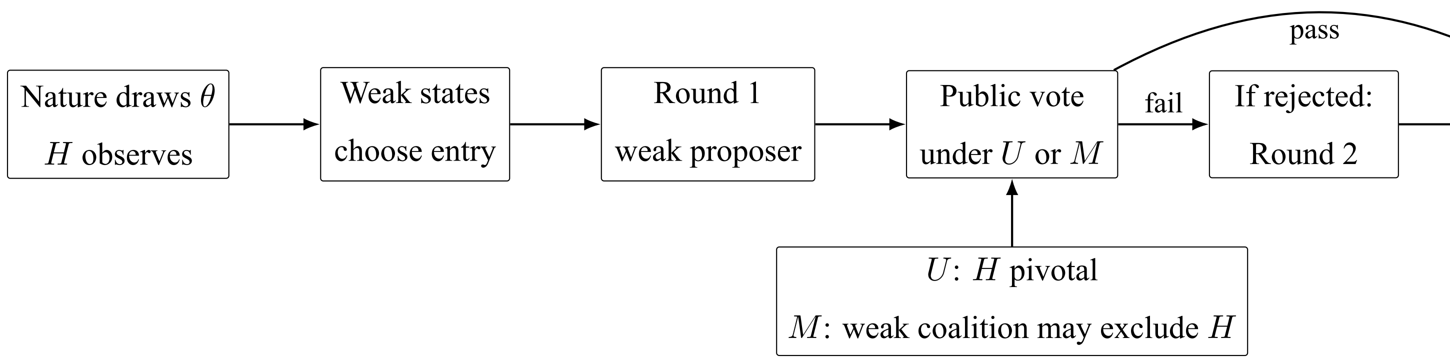

Figure 1: Baseline timing and institutional contrast. The hegemon has private information and is pivotal under unanimity. Under majority, the no-screening path allows weak states to pass a proposal without making the hegemon pivotal.

5 Majority Rule: The No-Screening Benchmark

Let

$$q = \lfloor N/2 \rfloor + 1, \qquad k = q - 1.$$

A recognized weak proposer is counted as voting yes, so it needs k additional yes votes under majority.

In terminal Round 2, a weak proposer can pass a majority proposal without H, set $y = 0$, and give zero to the required weak voters because terminal rejection gives weak states zero. The recognized weak proposer's terminal majority value is one, and the representative weak-state value is

$$W_2^M = \frac{1}{m}.$$

In Round 1, the continuation value of a weak voter after rejection is

$$c_M = \beta W_2^M = \frac{\beta}{m}.$$

The no-H majority proposal sets $y = 0$, gives c_M to k weak voters, and gives zero to everyone else. Its proposer payoff is $1 - kc_M$, and total weak payoff is one. Hence the representative weak-state entry payoff under majority is

$$V_W^M(\mu) = \frac{1}{m}. \tag{1}$$

Majority does not automatically eliminate screening. If buying the low-threshold hegemon is cheaper than buying a weak voter, a weak proposer may include H as a substitute coalition partner. Let a_0^M and a_1^M be the Round-1 majority thresholds of the low and high types if H is made pivotal in a majority coalition, with

$$a_\theta^M = t_\theta - o_\theta + \beta o_\theta = t_\theta - (1-\beta)o_\theta. \tag{Majority Dynamic Threshold}$$

The maintained Majority Threshold Order is

$$0 \leq a_0^M < a_1^M \leq \bar{y}. \tag{Majority Threshold Order}$$

Proposition 1 (Majority no-screening benchmark)**.** *Under the Majority Threshold Order, the no-H majority path weakly dominates all H-including majority proposals for every belief if and only if*

$$a_0^M \geq \frac{\beta}{m}. \tag{No-Cheap-H}$$

Equivalently, in primitives,

$$t_0 - (1-\beta)o_0 \geq \frac{\beta}{m}. \tag{No-Cheap-H primitives}$$

When No-Cheap-H holds, majority produces no screening of H: weak states pass a proposal without making H pivotal, $V_W^M(\mu) = 1/m$, and H's payoff is

$$V_H^M(\mu) = (1-\mu)o_0 + \mu o_1. \tag{2}$$

If $a_0^M < \beta/m$, majority may also screen H by buying the low type as a cheap coalition partner at sufficiently low beliefs.

Proof in Appendix A.1.

No-Cheap-H is the key scope condition for the paper's majority benchmark. Majority switches off pivotality-based informational power only when weak states do not want to substitute H's low-threshold approval for a weak voter's approval.

This benchmark is deliberately favorable to weak states. Majority without H preserves the entire weak coalition surplus, normalized to one. A natural extension would let a majority agreement without H generate only $\rho < 1$ units of weak surplus, yielding $V_W^M = \rho/m$. The baseline keeps $\rho = 1$ to isolate the informational consequence of making H's approval nonpivotal.

6 Unanimity: Pivotality and Passive Beliefs

Under unanimity, H's approval is necessary. The package y remains feasible in every state, but the proposer does not know which threshold it must satisfy.

6.1 Terminal Round

In Round 2, rejection ends bargaining. The proposer never benefits from offering a package strictly inside an acceptance region. The only relevant accepted packages are therefore $y = t_0$, accepted only by the low type, and $y = t_1$, accepted by both types. Let

$$p_2(\mu) = \max\{(1-\mu)(1-t_0),\, 1-t_1\} \tag{3}$$

be the recognized weak proposer's terminal value, with the low-threshold package selected at equality.

Lemma 1 (Terminal unanimity threshold)**.** *Under the Threshold Order, the Round-2 unanimity proposal is $y = t_0$ for*

$$\mu \leq \mu_2^* \equiv \frac{t_1 - t_0}{1 - t_0}, \tag{4}$$

and $y = t_1$ for $\mu > \mu_2^*$. The representative weak-state continuation value is

$$W_2^U(\mu) = \frac{p_2(\mu)}{m}. \tag{5}$$

Proof in Appendix A.2.

Let $C_0(\nu)$ and $C_1(\nu)$ denote H's Round-2 continuation payoffs after posterior ν. From Lemma 1,

$$C_1(\nu) = o_1, \qquad C_0(\nu) = \begin{cases} o_0, & \nu \leq \mu_2^*, \\ o_0 + t_1 - t_0, & \nu > \mu_2^*. \end{cases} \tag{6}$$

The low type gains an informational rent in Round 2 when the proposer pools at t_1.

6.2 Round 1

Let

$$c(\nu) = \beta W_2^U(\nu) = \frac{\beta p_2(\nu)}{m} \tag{7}$$

be a weak voter's continuation value after a failed Round-1 proposal that induces posterior ν. The Round-1 result is stated on the high-posterior pooling domain

$$\mu_2^* < 1, \qquad \text{equivalently} \qquad t_1 < 1, \tag{High-Posterior Pooling}$$

so posterior $\nu = 1$ strictly induces terminal pooling at t_1 and $C_0(1) = o_0 + t_1 - t_0$. The dynamic acceptance formula then gives the relevant Round-1 unanimity thresholds:

$$a_1 = t_1 - o_1 + \beta C_1(1) = t_1 - (1 - \beta) o_1, \tag{8}$$

and, when a low type's rejection is interpreted as the high posterior and therefore induces terminal pooling,

$$a_0^1 = t_0 - o_0 + \beta C_0(1) = t_0 - o_0 + \beta(o_0 + t_1 - t_0). \tag{9}$$

The R1 Dynamic Threshold Order is

$$0 \leq a_0^1 \leq a_1 \leq \bar{y} \leq 1. \tag{R1 Dynamic Threshold Order}$$

Strict low-only separation requires $a_0^1 < a_1$. The inequality $a_0^1 \leq a_1$ is a substantive single-crossing restriction, not a bookkeeping detail. It requires the low type's value from rejecting into a high-posterior continuation not to overturn the terminal threshold order.

Under the passive-belief assessment, the weak proposer faces exactly three payoff-relevant candidates: insure agreement, test the low threshold, or wait.

Pooling P. The proposer offers $y = a_1$ and pays each non-proposing weak voter $c(\mu)$. Both types of H accept. The proposer payoff is

$$\Pi_P^U(\mu) = 1 - a_1 - (m-1)c(\mu), \tag{10}$$

provided $a_1 + (m-1)c(\mu) \leq 1$.

Low-only L. The proposer offers $y = a_0^1$ and pays each non-proposing weak voter $c(0)$. The low type accepts, the high type rejects, and a high-type rejection reveals posterior one. The proposer payoff is

$$\Pi_L^U(\mu) = (1-\mu)\{1 - a_0^1 - (m-1)c(0)\} + \mu c(1), \tag{11}$$

provided $a_0^1 + (m-1)c(0) \leq 1$ and $a_0^1 < a_1$.

No-information rejection R. The proposer offers $y = 0$, assigns zero to every weak state, and designates one weak voter to reject. The proposal is designed to fail because the designated weak voter rejects. All other weak voters and both types of H are prescribed to vote yes; if H unexpectedly changes its vote, the ballot still fails by the weak rejection and the assessment keeps posterior μ. The proposer payoff is

$$\Pi_R^U(\mu) = c(\mu). \tag{12}$$

This candidate is always feasible because it uses no positive allocation from the fixed pie.

Let $\mathcal{K}(\mu) \subseteq \{P, L, R\}$ collect candidates satisfying their feasibility and separation conditions:

$$P \in \mathcal{K}(\mu) \iff a_1 + (m-1)c(\mu) \leq 1,$$

$$L \in \mathcal{K}(\mu) \iff a_0^1 + (m-1)c(0) \leq 1 \quad \text{and} \quad a_0^1 < a_1,$$

and

$$R \in \mathcal{K}(\mu) \quad \text{always.}$$

Proposition 2 (R1 outcome under the passive-belief assessment)**.** *Under the Threshold Order, High-Posterior Pooling, the R1 Dynamic Threshold Order, simultaneous public voting, the passive-belief assessment in Definition 2, weak approval at pivotal constraints, and the H-payoff-minimizing tie-break among weak-proposer payoff ties, the selected Round-1 unanimity candidate lies in $\{P, L, R\}$. The selected weak-proposer value is*

$$\bar{\Pi}^U(\mu) = \max_{K \in \mathcal{K}(\mu)} \Pi_K^U(\mu). \tag{13}$$

This is a candidate-comparison result under the maintained passive-belief selection rule. It does not construct a full strategy profile after every arbitrary off-path proposal and does not characterize unrestricted PBEs.

Proof in Appendix A.3.

Remark 1 (Delay is endogenous)**.** *The rejection candidate R is not imposed by assumption. It is a feasible candidate under the specified passive assessment when continuation is more valuable to the weak proposer than currently available accepted packages. Thus delay can arise from incentives, but it is not required for the calibrated result below.*

7 Entry, Nesting, and Institutional Choice

Entry by weak states is collective. The entry payoff must therefore count the total weak payoff generated by the selected path, not only the residual held by the recognized proposer. If $k^*(\mu) \in$

$\{P, L, R\}$ is the selected unanimity candidate from Proposition 2, define total weak payoff under each candidate as

$$S_P^U(\mu) = 1 - a_1, \tag{14}$$

$$S_L^U(\mu) = (1-\mu)(1-a_0^1) + \mu\beta p_2(1), \tag{15}$$

and

$$S_R^U(\mu) = \beta p_2(\mu). \tag{16}$$

The representative weak-state entry payoff under unanimity is

$$V_W^U(\mu) = \frac{S_{k^*(\mu)}^U(\mu)}{m}. \tag{17}$$

For an entry cost $\chi \geq 0$, define

$$F_R(\chi) = \{\mu \in [0,1] : V_W^R(\mu) \geq \chi\}, \qquad R \in \{U, M\}. \tag{18}$$

Proposition 3 (Weak-state entry nesting)**.** *Under the threshold domains above, High-Posterior Pooling, and the No-Cheap-H condition,*

$$V_W^U(\mu) \leq V_W^M(\mu) = \frac{1}{m} \quad \textit{for every } \mu \in [0,1]. \tag{19}$$

Consequently,

$$F_U(\chi) \subseteq F_M(\chi) \quad \textit{for every } \chi \geq 0. \tag{20}$$

Proof in Appendix A.4.

Nesting means that majority can form whenever unanimity can form, but not conversely. Under the fixed-pie benchmark this result is close to accounting: majority lets the weak coalition keep the full unit surplus, while unanimity either gives part of the surplus to H or delays agreement. The substantive comparison is therefore not nesting itself, but the trade-off between weaker entry incentives and the hegemon's payoff gap $\Delta_H(\mu)$.

The hegemon's unanimity payoff under the selected candidate is

$$V_H^U(\mu) = H_{k^*(\mu)}(\mu), \tag{21}$$

where

$$H_P(\mu) = (1-\mu)(o_0 + a_1 - t_0) + \mu(o_1 + a_1 - t_1), \tag{22}$$

$$H_L(\mu) = (1-\mu)(o_0 + a_0^1 - t_0) + \mu\beta C_1(1), \tag{23}$$

and

$$H_R(\mu) = (1-\mu)\beta C_0(\mu) + \mu\beta C_1(\mu). \tag{24}$$

Define the hegemon's institutional payoff gap as

$$\Delta_H(\mu) = V_H^U(\mu) - V_H^M(\mu). \tag{25}$$

Proposition 4 (Conditional comparison for the hegemon)**.** *On beliefs where both institutions form, H's ranking is determined by the sign of $\Delta_H(\mu)$. It strictly prefers unanimity iff $\Delta_H(\mu) > 0$, is indifferent iff $\Delta_H(\mu) = 0$, and strictly prefers majority iff $\Delta_H(\mu) < 0$.*

Proof in Appendix A.5.

Corollary 1 (Institutional classification under the baseline assessment)**.** *Under the conditions of Proposition 3, the following five sets form a mutually exclusive and exhaustive partition of $[0, 1]$:*

$$\mathcal{C}_0(\chi) = [0, 1] \setminus F_M(\chi), \tag{26}$$

$$\mathcal{C}_M(\chi) = F_M(\chi) \setminus F_U(\chi), \tag{27}$$

$$\mathcal{C}_U^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) > 0\}, \tag{28}$$

$$\mathcal{C}_T^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) = 0\}, \tag{29}$$

and

$$\mathcal{C}_M^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) < 0\}. \tag{30}$$

They correspond, respectively, to no institution forming; only majority forming; both forming with H preferring unanimity; both forming with H indifferent; and both forming with H preferring majority.

Proof in Appendix A.5.

The classification is deliberately weaker than an unconditional ranking theorem. Unanimity can benefit the hegemon through pivotality-based screening, but majority can be better when the screening rent is smaller than the value of exclusion under the majority benchmark.

8 Working Calibration

Example 1 (Working calibration)**.** *Set $N = 13$, $m = 12$, $\beta = 0.9$, $\bar{y} = 1$, $t_0 = 0.19$, and $t_1 = 0.285$. For the payoff comparison, set $o_0 = t_0$ and $o_1 = t_1$. The implied values are*

$$\mu_2^* = 0.117283950617, \qquad a_1 = 0.2565, \qquad a_0^1 = 0.2565,$$

so High-Posterior Pooling holds and the calibration lies on the R1 Dynamic Threshold Order boundary $a_0^1 = a_1$. Strict low-only separation is therefore blocked in the calibration, and the selected Round-1 unanimity candidate is pooling for all beliefs. Majority satisfies No-Cheap-H because

$$a_0^M = 0.171 > \frac{\beta}{m} = 0.075.$$

The weak-state entry payoffs are

$$V_W^U(\mu) = \frac{1 - 0.2565}{12} = 0.0619583, \qquad V_W^M(\mu) = \frac{1}{12} = 0.0833333.$$

The hegemon's payoff gap is

$$\Delta_H(\mu) = 0.0665 - 0.095\mu. \tag{31}$$

Thus, when both rules form, H prefers unanimity for $\mu < 0.7$, is indifferent at $\mu = 0.7$, and prefers majority for $\mu > 0.7$.

Condition	Value	Boundary	Margin
No-Cheap-H	$a_0^M = 0.171$	$\beta/m = 0.075$	0.096
$t_0 \geq 0$	0.190	0	0.190
$t_1 \geq t_0$	0.285	0.190	0.095
$\bar{y} \geq t_1$	1.000	0.285	0.715
$1 \geq \bar{y}$	1.000	1.000	0.000

Table 1: Calibration margins for No-Cheap-H and the terminal Threshold Order. The zero slack in the last row is the normalization $\bar{y} = 1$.

R1 comparison	Value	Boundary or rival	Margin/status
P feasibility at max $c(\mu)$	0.924750	1	0.075250
R feasibility	0.000000	1	1.000000
$a_1 - a_0^1$	0.000	> 0 for strict L	Boundary; strict L blocked
$\Pi_P^U(0) - \Pi_R^U(0)$	$0.075250 - 0.060750$	0	0.014500
$\Pi_P^U(\mu_2^*) - \Pi_R^U(\mu_2^*)$	$0.153625 - 0.053625$	0	0.100000
$\Pi_P^U(0.5) - \Pi_R^U(0.5)$	$0.153625 - 0.053625$	0	0.100000
$\Pi_P^U(1) - \Pi_R^U(1)$	$0.153625 - 0.053625$	0	0.100000

Table 2: Round-1 calibration margins under the passive-belief assessment. The equality $a_0^1 = a_1$ places the low-only candidate on the strict-separation boundary, while pooling beats no-information rejection at the displayed beliefs.

Object	Value	Interpretation
V_W^M	0.083333	Representative weak payoff under majority
V_W^U	0.061958	Representative weak payoff under selected unanimity path
$V_W^M - V_W^U$	0.021375	Entry/nesting gap

Table 3: Entry and nesting margin in the working calibration.

Belief μ	$\Delta_H(\mu) = V_H^U(\mu) - V_H^M(\mu)$
0	0.066500
0.5	0.019000
0.7	0.000000
1	-0.028500

Table 4: Hegemon payoff-gap margins in the working calibration. Positive values favor unanimity; negative values favor majority.

Entry cost per weak state χ	Institutional outcome
$\chi \leq 0.0619583$	Both rules form; H prefers unanimity for $\mu < 0.7$
$0.0619583 < \chi \leq 0.0833333$	Only majority forms
$\chi > 0.0833333$	No rule forms

Table 5: Calibrated institutional classification under the fixed-pie relative-package baseline. At $\mu = 0.7$, H is indifferent when both rules form; for $\mu > 0.7$, H prefers majority.

9 Discussion

9.1 Illustration: OPEC and Saudi pivotality

OPEC is a useful illustration because the institutional package is naturally relative. Members bargain over quota shares, production ceilings, monitoring, flexibility, exceptions, and burden sharing. These are not cash transfers that must fit a different realized pie in each state. They are institutional terms that shift the residual benefits of cooperation across members. A larger quota share or a weaker cut obligation for Saudi Arabia is a higher y: it is better for the pivotal producer and costlier for the rest of the coalition.

Model object	OPEC interpretation
H	Saudi Arabia as privately informed pivotal producer
Weak states	Other OPEC members or non-hegemonic bargaining coalition
y	Quota, flexibility, exception, monitoring, or enforcement package favoring Saudi Arabia
t_θ	Saudi net participation threshold for accepting restraint terms
μ	Belief that Saudi Arabia is in the high-threshold state
χ	Formation, participation, compliance, or coordination cost for weak members
$\pi_H = 0$	Baseline with no formal Saudi proposal recognition

Table 6: Mapping between the formal baseline and the OPEC illustration.

In this interpretation, Saudi Arabia is the privately informed pivotal participant, not the formal agenda setter in the baseline model. The relevant private information concerns its net participation threshold: how attractive an OPEC package must be relative to unilateral production, spare-capacity deployment, price-war threats, and the domestic and fiscal costs of restraint. Other members observe the proposed quota package, but they do not directly observe the threshold at which Saudi Arabia is willing to keep restraining production. That is the screening problem.

The historical record fits the informational premise. Saudi Arabia has long been treated as the central swing producer in oil-market accounts, with unusually large low-cost capacity and a distinctive ability to discipline cartel outcomes (Alhajji and Huettner 2000; Nakov and Nuño 2013; Fattouh and Mahadeva 2013). The 1985–86 price collapse illustrates why other members cared about Saudi participation: a breakdown in restraint could produce severe collective losses (Griffin and Neilson 1994; Yergin 1991). At the same time, the exact cost and credibility of Saudi alternatives were hard for other members to verify. Analyses of Saudi reserves and spare capacity emphasize persistent uncertainty about the true usable capacity behind official claims (Simmons 2005; Fattouh and Mahadeva 2013).

The model's point is not that Saudi Arabia writes every OPEC package. The point is sharper: when OPEC decisions require consensus, a package that cannot secure Saudi approval fails. Even if weak or smaller producers formally control the agenda, they must choose terms that satisfy a privately known Saudi threshold. Majority rule would alter that informational constraint if a coalition of other members could set terms without making Saudi approval pivotal. Whether such a majority rule would be enforceable in the real organization is a separate institutional question; the model isolates the distributive consequence of pivotality.

The working calibration should be read in this spirit. The parameters $t_0 = 0.19$ and $t_1 = 0.285$ are not direct estimates of Saudi payoffs. They describe a threshold gap large enough to make pivotal approval informationally valuable, while retaining a fixed weak surplus and no formal proposal advantage for H. Under this calibration, unanimity forms only when $\chi \leq 0.0619583$, majority forms up to $\chi = 0.0833333$, and when both form the pivotal actor prefers unanimity for $\mu < 0.7$. The classification therefore replaces a simple "consensus always helps the hegemon" claim with a more precise prediction: consensus helps the hegemon only where the pivotality-screening rent exceeds the payoff from being excluded under majority.

9.2 Scope

The baseline intentionally stacks agenda power against the hegemon by setting $\pi_H = 0$. This is not a claim that powerful states never influence agendas in real organizations. It is a modeling discipline. If unanimity can benefit a privately informed hegemon even when only weak states are recognized to make proposals, then the mechanism is not reducible to agenda control.

No-Cheap-H is equally important. Majority removes screening only when a weak proposer prefers buying weak-state votes to buying the low-threshold hegemon as a cheap coalition partner. When $a_0^M < \beta/m$, majority may also create screening. That extension is substantively plausible in small committees or settings where the pivotal powerful actor is very cheap to satisfy in one state. It is not part of the baseline because the paper's main contrast is the case in which majority lets weak states bypass the private threshold altogether.

The passive-belief restriction is also a scope condition. It is natural here because weak states have no private information about H's type. Their unexpected votes should not be interpreted as signals about information they do not possess. But this is a maintained equilibrium assessment, not a theorem about every unrestricted PBE. A different belief discipline could support additional off-path behavior, especially after histories that the baseline treats as uninformative. The paper's formal claim is therefore a clean institutional classification under a stated voting and belief protocol.

Finally, the binary type space is a device for isolating the mechanism. The threshold gap $t_1 - t_0$ is the primitive source of informational power. Continuous types, heterogeneous weak states, learning across repeated OPEC meetings, and positive recognition probability for H are natural extensions. They should be added only after preserving the separation among outside-option payoffs, veto or pivotality power, and proposal power that the current architecture enforces.

10 Conclusion

Consensus is usually described as a constraint on power. This paper shows why that description is incomplete. When a powerful state is privately informed and its approval is necessary, consensus can turn a veto into an informational asset. Weak states must choose how generous a package to offer without knowing the hegemon's true participation threshold. The informational rent comes from pivotality, not from formal proposal power.

The fixed-pie relative-package model separates this mechanism from two nearby sources of power. Majority rule can remove screening when weak states can form a winning coalition without H and when H is not a cheaper coalition partner than a weak voter. Unanimity creates screening because every package must satisfy H's private threshold. Agenda power is held out of the baseline by setting $\pi_H = 0$. The central result is therefore not that unanimity globally dominates majority, but that the institutional ranking can be classified exactly from two objects: the weak-state entry sets and the sign of $\Delta_H(\mu)$.

This distinction matters empirically. In organizations such as OPEC, the relevant bargaining terms are quota shares, production flexibility, and enforcement burdens rather than literal transfers. Those terms can make a pivotal privately informed actor better off even when smaller members formally participate as equals. At the same time, the model predicts limits. If entry costs are too high, unanimity may not form. If beliefs make the private threshold less valuable, majority can be better for H when both rules are viable.

The broader implication is that formal equality and asymmetric power can coexist through information. Consensus does not merely protect weak states from domination, and majority does not always favor the powerful. Which rule benefits a hegemon depends on whether the voting rule makes its private participation threshold politically productive.

Appendix A: Proofs of Main Results

A.1 Majority no-screening

Proof of Proposition 1. In terminal majority bargaining, a weak proposer can pass without H, set $y = 0$, pay zero to the required weak voters, and keep the unit surplus. Hence $W_2^M = 1/m$, and a non-proposing weak voter's Round-1 continuation value is $c_M = \beta/m$. In Round 1, the no-H proposal sets $y = 0$, pays c_M to $k = q - 1$ weak voters, pays zero to all other weak states, and keeps $1 - kc_M$ for the proposer. Since $q \leq m$ for $N \geq 3$ and $\beta \leq 1$,

$$1 - kc_M - c_M = 1 - \frac{q\beta}{m} \geq 0,$$

so the accepted no-H proposal weakly dominates deliberate rejection.

Any H-including majority proposal that also buys k weak votes is weakly dominated by dropping H, because H's vote is not needed. The only relevant H-including alternatives buy H and $k - 1$ weak voters. Pooling with H costs $a_1^M + (k-1)c_M$, so it cannot beat the no-H proposal when $a_1^M \geq c_M$. A low-only H-including proposal gives the weak proposer

$$(1 - \mu)\{1 - a_0^M - (k-1)c_M\} + \mu c_M.$$

Relative to the no-H proposal, the payoff gap is

$$c_M - a_0^M + \mu(a_0^M + kc_M - 1). \tag{A1}$$

If $a_0^M \geq c_M$, this gap is nonpositive at $\mu = 0$. At $\mu = 1$, it equals $(k+1)c_M - 1 = q\beta/m - 1 \leq 0$. Since (A1) is affine in μ, the low-only H-including proposal is weakly dominated for every μ. Because $a_1^M > a_0^M$, the same condition implies $a_1^M \geq c_M$, so pooling with H is also not profitable.

Necessity follows by evaluating (A1) at $\mu = 0$. If $a_0^M < c_M$, the low-only H-including proposal strictly beats the no-H proposal at $\mu = 0$ and therefore for nearby beliefs. The cutoff for this majority-screening region is

$$\mu_M^H = \frac{c_M - a_0^M}{1 - a_0^M - kc_M},$$

whenever $a_0^M < c_M$. This proves the iff statement and the extension claim. □

A.2 Terminal unanimity

Proof of Lemma 1. In terminal Round 2, a package below t_0 is rejected by both types and gives weak states zero. A package in $[t_0, t_1)$ is accepted only by the low type; within this region, $y = t_0$ maximizes weak surplus because any higher y reduces the fixed weak surplus one-for-one. A package at least t_1 is accepted by both types; within this region, $y = t_1$ maximizes weak surplus. Terminal weak voters have zero continuation value, so the proposer need not allocate positive x_i's to them. Thus the weak proposer compares

$$L_2(\mu) = (1 - \mu)(1 - t_0)$$

with

$$P_2(\mu) = 1 - t_1.$$

The low-threshold package is weakly better iff

$$(1 - \mu)(1 - t_0) \geq 1 - t_1,$$

which is equivalent to $\mu \leq (t_1 - t_0)/(1 - t_0)$. Dividing the selected proposer value by m gives the representative weak-state value. □

A.3 R1 passive-belief unanimity

Proof of Proposition 2. Fix a belief μ and the passive-belief assessment in Definition 2.

Step 1: Candidate exhaustion. Consider accepted Round-1 unanimity proposals. Under the R1 Dynamic Threshold Order, if a package is high enough for the high type under the relevant continuation, it is also high enough for the low type. Thus high-only acceptance is ruled out. Any accepted package strictly above the relevant dynamic threshold is weakly dominated for the proposer by lowering y, and any weak-voter allocation strictly above the relevant continuation value is weakly dominated by lowering that allocation. Therefore the only payoff-relevant accepted candidates are pooling P and low-only L. Rejected proposals either preserve the prior, in which case they are payoff-equivalent to R, or try to reveal information through H's vote. Step 5 rules out the latter as a payoff-improving fourth candidate.

Step 2: Pooling assessment. For P, the proposal sets $y = a_1$, pays every non-proposing weak voter $c(\mu)$, and assigns the residual to the proposer. Both types of H vote yes. The high type is indifferent at $a_1 = t_1 - o_1 + \beta C_1(1)$. The low type accepts because $a_1 \geq a_0^1$. A non-proposing weak voter who votes yes receives $c(\mu)$; if it deviates to no and causes failure, H's yes vote pools types and the posterior remains μ, so the deviation payoff is also $c(\mu)$. Hence weak voters are willing to approve. The proposer payoff is

$$1 - a_1 - (m-1)c(\mu) = \Pi_P^U(\mu).$$

Step 3: Low-only assessment. For L, the proposal sets $y = a_0^1$, pays every non-proposing weak voter $c(0)$, and assigns the residual to the proposer. The low type votes yes at its dynamic threshold. The high type votes no when $a_0^1 < a_1$, inducing posterior one. A non-proposing weak voter's approval incentive in the low state is

$$(1-\mu)x + \mu c(1) \geq (1-\mu)c(0) + \mu c(1),$$

which is equivalent to $x \geq c(0)$. Thus the minimal weak-voter allocation is $x = c(0)$. The proposer receives the accepted residual in the low state and the continuation value $c(1)$ in the high state, giving

$$(1-\mu)\{1 - a_0^1 - (m-1)c(0)\} + \mu c(1) = \Pi_L^U(\mu).$$

Step 4: Rejection assessment. For R, the proposer sets $y = 0$, assigns zero to every weak state, names a designated weak voter W_j, and prescribes that W_j vote no. This proposal is feasible because $y + \sum_i x_i = 0$. All other weak voters and both types of H are prescribed to vote yes. Given W_j's rejection, their votes are nonpivotal: either vote leaves the ballot failed, and Definition 2 keeps posterior μ. Thus they receive the same continuation payoff after either vote. If W_j follows the prescription, it receives the continuation value $c(\mu)$. If W_j deviates to yes while the other voters follow their prescriptions, the proposal passes and W_j receives zero. Thus the designated weak rejecter has no profitable deviation. The proposer payoff is $c(\mu) = \Pi_R^U(\mu)$.

Step 5: Informative rejected ballots. An informative rejected ballot would require H's vote to separate types while the proposal fails. Under High-Posterior Pooling, the low type's discounted continuation from inducing posterior one is

$$\beta C_0(1) = \beta(o_0 + t_1 - t_0),$$

whereas revealing the low posterior gives

$$\beta C_0(0) = \beta o_0.$$

Since $t_1 > t_0$ and $\beta > 0$,

$$\beta C_0(1) > \beta C_0(0),$$

so the low type strictly prefers mimicking the high-posterior rejected vote to revealing the low posterior. Separating rejected ballots are therefore not incentive compatible for the low type. Pooling rejected ballots preserve the prior and are payoff-equivalent to R. On the excluded boundary $\mu_2^* = 1$, this comparison can bind; the paper does not use that boundary for the R1 candidate-comparison result.

Step 6: Proposer optimality and tie-break. The admissible set $\mathcal{K}(\mu)$ contains the feasible candidates among P, L, R. By Steps 1–5, no other accepted or rejected ballot yields a higher weak-proposer payoff under the stated passive-belief candidate comparison. The weak proposer selects a payoff maximizer in $\mathcal{K}(\mu)$, and ties are broken by minimizing H's expected payoff. Thus the selected candidate lies in $\{P, L, R\}$ under the maintained passive-belief selection rule. □

A.4 Entry nesting

Proof of Proposition 3. Under pooling, total weak payoff is $S_P^U = 1 - a_1 \leq 1$. Under low-only,

$$S_L^U(\mu) = (1-\mu)(1-a_0^1) + \mu\beta p_2(1) \leq (1-\mu) + \mu = 1,$$

because $a_0^1 \geq 0$, $p_2(1) \leq 1$, and $\beta \leq 1$. Under rejection,

$$S_R^U(\mu) = \beta p_2(\mu) \leq 1.$$

The selected unanimity candidate is one of these three. Hence $S^U_{k^*(\mu)}(\mu) \leq 1$ and $V_W^U(\mu) \leq 1/m$. Proposition 1 gives $V_W^M(\mu) = 1/m$ under No-Cheap-H. Therefore $F_U(\chi) \subseteq F_M(\chi)$ for every χ. □

A.5 Conditional comparison and classification

Proof of Proposition 4. On $F_U(\chi)$, both rules form. By definition, $V_H^U(\mu) - V_H^M(\mu) = \Delta_H(\mu)$. The sign of the difference is exactly the sign of the institutional preference. □

Proof of Corollary 1. The five sets split the belief space first by whether majority forms, then by whether unanimity also forms, and finally by the sign of $\Delta_H(\mu)$ on the both-form region. These categories are disjoint and exhaustive by construction. Proposition 3 rules out an only-unanimity category. □

Appendix B: Protocol, Payoffs, and Scope

B.1 Passive beliefs

The passive-belief refinement is the assessment stated in Definition 2. Its role is limited. It specifies how public ballots update beliefs when weak voters deviate, when H's vote separates types, when H deviates from a pooling yes or pooling no prescription, and when H's vote changes in a proposal already designed to fail by weak rejection. These restrictions are part of the maintained solution concept used in the baseline and are not presented as implications of unrestricted PBE.

The tie-break against H does not select beliefs and does not resolve off-path multiplicity. It only selects among weak-proposer payoff-maximizing candidates after the passive assessment has already determined which candidates are admissible.

B.2 Threshold microfoundation

The body uses t_θ as the primitive net threshold. One can microfound it by writing d_θ for H's external payoff and b_θ for the direct benefit of agreement apart from the package. Then H accepts terminal package y iff

$$y + b_\theta \geq d_\theta,$$

so

$$t_\theta = d_\theta - b_\theta. \tag{B1}$$

All main results depend on the net thresholds and the continuation thresholds a_0^1, a_1, a_0^M, a_1^M, not on a separate decomposition into d_θ and b_θ.

B.3 Extensions not used in the baseline

If $a_0^M < \beta/m$, majority may also screen H. In that case, the no-H benchmark is no longer the right majority comparison because the low type of H is cheaper than a weak voter. This is an extension, not a contradiction of the mechanism. It identifies the domain in which majority truly removes H's pivotality.

If $\pi_H > 0$, agenda power becomes a separate channel. The present baseline sets $\pi_H = 0$ to isolate informational power through pivotality. A model with positive recognition probability for H should be treated as an extension because it combines pivotality with agenda power.

Continuous type spaces are also left for future work. The binary model is used to make the pivotality-screening mechanism and the institutional classification transparent under a fully checked fixed-pie protocol.

Appendix C: Notation and Reproducibility

C.1 Notation

Symbol	Type	Meaning	First use / Used in
N	Primitive	Total number of states	Definition 1
$m = N - 1$	Derived primitive	Number of weak states	Definition 1
q	Rule parameter	Majority quota, $q = \lfloor N/2 \rfloor + 1$	Majority benchmark
$k = q - 1$	Derived rule parameter	Additional yes votes a weak proposer needs under majority	Majority benchmark
θ	Type	Hegemon type, low 0 or high 1	Definition 1
μ	Belief	Prior probability that $\theta = 1$	Definition 1
y	Choice variable	Relative institutional package favoring H	Definition 1
t_θ	Primitive threshold	Terminal net participation threshold of type θ	Definition 1
$\bar{y}$	Primitive bound	Maximum admissible package	Definition 1
β	Primitive	Common discount factor	Definition 1
π_H	Recognition parameter	Recognition probability of H; baseline sets $\pi_H = 0$	Definition 1
χ	Entry cost	Cost per weak state of institutional entry	Entry section
o_θ	Primitive payoff	External payoff of type θ when no agreement includes H	Definition 1
a_0^M, a_1^M	Derived thresholds	Round-1 majority thresholds if H is made pivotal	Proposition 1
a_0^1, a_1	Derived thresholds	Round-1 unanimity thresholds under high-posterior continuation	Proposition 2
$p_2(\mu)$	Derived value	Terminal weak-proposer value under unanimity	Lemma 1
$c(\nu)$	Derived value	Weak voter's discounted continuation value after failed R1 ballot	Proposition 2
P, L, R	Candidate outcomes	Pooling, low-only, and no-information rejection candidates	Proposition 2
F_U, F_M	Entry sets	Beliefs for which unanimity or majority forms at cost χ	Proposition 3
Δ_H	Payoff gap	H's unanimity payoff minus majority payoff	Proposition 4

C.2 Reproducibility ledger

The formal results transported here are checked by separate R scripts:

Script	Object checked
`scripts/verify_relative_package_R2_piH0.R`	Terminal unanimity and μ_2^*
`scripts/verify_relative_package_R1_piH0.R`	R1 candidates, ICs, and tie-break selectio
`scripts/verify_relative_package_majority_piH0.R`	Majority no-screening iff $a_0^M \geq \beta/m$
`scripts/verify_relative_package_entry_nesting_piH0.R`	Weak-state nesting and $\Delta_H(\mu)$
`scripts/verify_relative_package_classification_piH0.R`	Five-way classification and calibration
`scripts/verify_relative_package_margins_piH0.R`	Calibration margin tables

Table 8: Reproducibility ledger for the fixed-pie relative-package $\pi_H = 0$ baseline.

The working calibration reported in Example 1 is reproduced by the classification script. It verifies $V_W^U = 0.0619583$, $V_W^M = 0.0833333$, minimum weak-state nesting gap 0.021375, and $\Delta_H(\mu) = 0.0665 - 0.095\mu$.

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